Maternal Vitamin D, Obesity and Immune Gene Expression in Preeclampsia

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1 Methods

Data source. Data came from a nested case-control study within the Vitamin D Antenatal Asthma Reduction Trial (VDAART) [1], available in the Gene Expression Omnibus as GSE85307. Peripheral whole-blood gene expression was measured at 10–18 weeks of gestation with the Affymetrix Human Gene 1.0 ST Array; values are on the $\log_2$ scale (range 3.1–13.2). The initial data set contained 157 samples and 500 probes. Covariates were matched to the expression data by GEO sample ID, and all 157 samples matched. Probe IDs were linked to gene symbols with the provided linker file.

Final sample size. No samples had missing values for the variables analyzed, so no data points were removed. The final sample size was 157 women: 110 with normal pregnancies and 47 who developed preeclampsia.

Variables. The genes of interest were *LYN*, *JAK2* and *PTX3*, chosen because preeclampsia involves maternal immune and inflammatory changes. The continuous covariate of interest was maternal 25-hydroxyvitamin D (ng/mL). The categorical covariates were pregnancy outcome (normal vs. preeclampsia) and maternal obesity status (not obese, BMI < 30 kg/m^2; obese, BMI ≥ 30 kg/m^2), derived from continuous BMI. Maternal age and gestational age at delivery were included as two additional continuous covariates in the summary table.

Analyses. Continuous covariates were summarized as mean (standard deviation) and categorical covariates as n (%), stratified by pregnancy outcome. The distribution of *LYN* expression was shown with a histogram, its association with vitamin D was measured with a Pearson correlation and a linear fit, and its expression by pregnancy outcome and obesity status was shown with boxplots. Ten genes (*LYN*, *JAK2*, *PTX3* and the seven genes with the highest standard deviation of expression across all women among the 500 probes) were shown in a heatmap of row-scaled (z-score) expression, with hierarchical clustering of genes and samples (Euclidean distance, complete linkage). Finally, a dumbbell plot compared the mean z-score of the same ten genes between pregnancy outcomes. All analyses were descriptive.

Software. All analyses were performed in R version 4.5.3 [2] with the packages ggplot2 (v4.0.3) [3], ggpubr (v1.0.0) [4], pheatmap (v1.0.13) [5] and knitr (v1.52) [6].

Table 1: Characteristics of the study participants by pregnancy outcome. Continuous variables are mean (standard deviation); categorical variables are n (%).

Variable	Normal (n = 110)	Preeclampsia (n = 47)	Overall (n = 157)
Maternal vitamin D (ng/mL)	24.4 (14.5)	19.7 (8.3)	23.0 (13.1)
Maternal age (years)	26.6 (5.1)	25.8 (4.9)	26.4 (5.0)
Gestational age at delivery (weeks)	39.3 (1.0)	37.8 (3.5)	38.8 (2.2)
Obesity status, n (%)			
Not obese (BMI < 30)	79 (71.8)	21 (44.7)	100 (63.7)
Obese (BMI ≥ 30)	31 (28.2)	26 (55.3)	57 (36.3)

Code availability. All code is available at `https://github.com/cheonijin927/QBS103-Final-Project`.

2 Results

2.1 Participant characteristics

Women who developed preeclampsia had lower mean vitamin D levels than women with normal pregnancies (19.7 vs. 24.4 ng/mL), delivered earlier on average (37.8 vs. 39.3 weeks) and were more often obese (55.3% vs. 28.2%), while maternal age was similar between groups (25.8 vs. 26.6 years) (Table 1).

2.2 *LYN* expression

LYN expression was approximately symmetric around a mean of 12.18 $\log_2$ units (Figure 1A). *LYN* expression was positively associated with maternal vitamin D (Pearson $r = 0.39$, $p < 0.001$; Figure 1B). Within each pregnancy outcome, median *LYN* expression was slightly lower in obese than in non-obese women (normal: 12.05 vs. 12.26; preeclampsia: 12.23 vs. 12.30), but the differences were small and the groups overlapped substantially (Figure 1C).

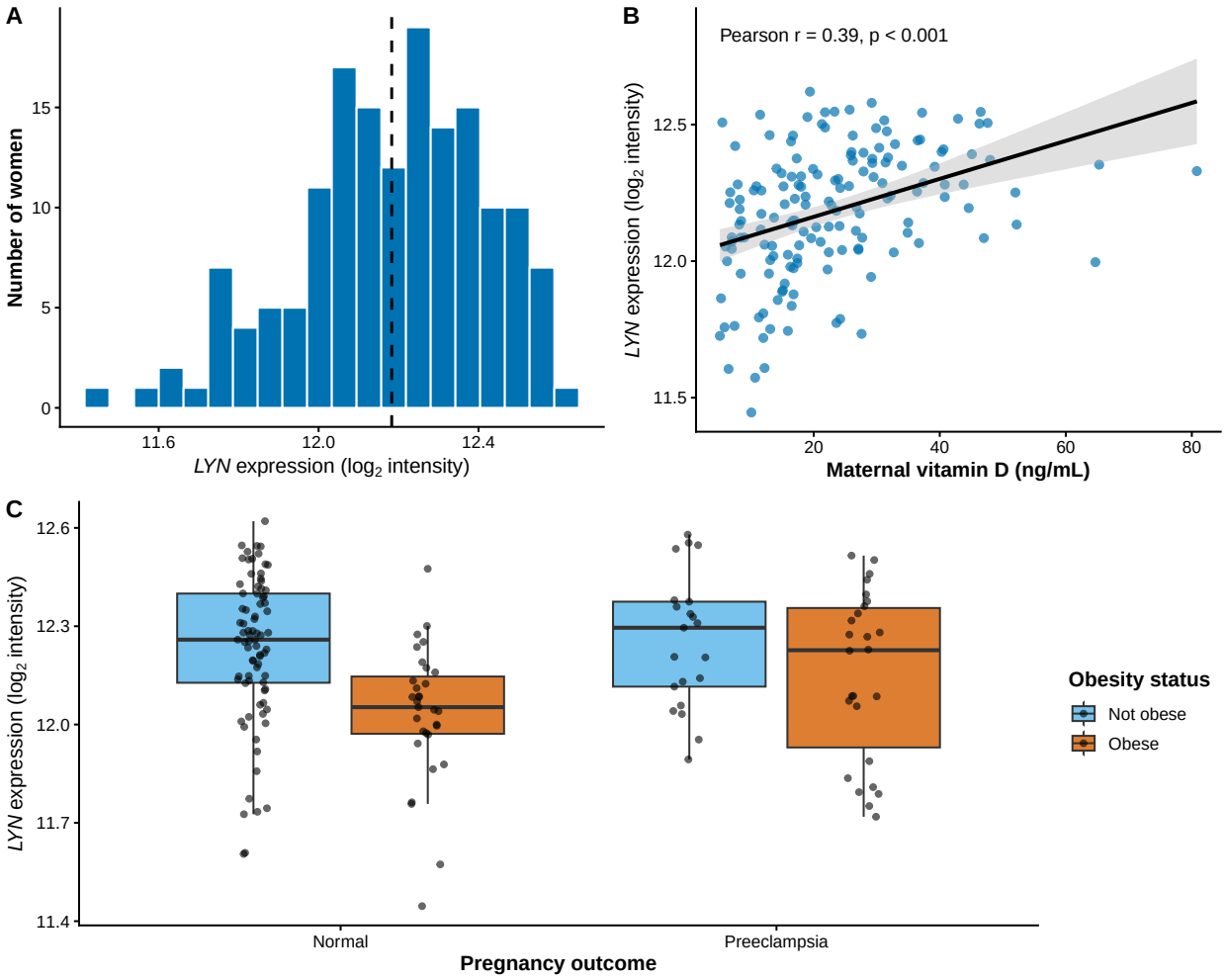


Figure 1: *LYN* expression in whole-blood samples from 157 pregnant women. (A) Distribution of *LYN* expression ($\log_2$ intensity); the dashed line marks the mean. (B) *LYN* expression vs. maternal vitamin D (25-hydroxyvitamin D, ng/mL); each point is one woman, the line is the linear fit with its 95% confidence band, and the Pearson correlation is shown in the upper left. (C) *LYN* expression by pregnancy outcome and obesity status (not obese, BMI < 30 kg/m^2; obese, BMI ≥ 30 kg/m^2). Boxes show the median and interquartile range, whiskers extend to 1.5 times the interquartile range, and points are individual women.

2.3 Heatmap of ten genes

The ten genes shown were *LYN*, *JAK2*, *PTX3*, *UTS2*, *RUNDC3A*, *TMEM176B*, *CEACAM6*, *CLIC2*, *SLC1A5* and *ZNF594*. Clustering grouped genes by their shared expression pattern across women, while women with preeclampsia and obese women were spread across the sample clusters rather than forming a separate group (Figure 2).

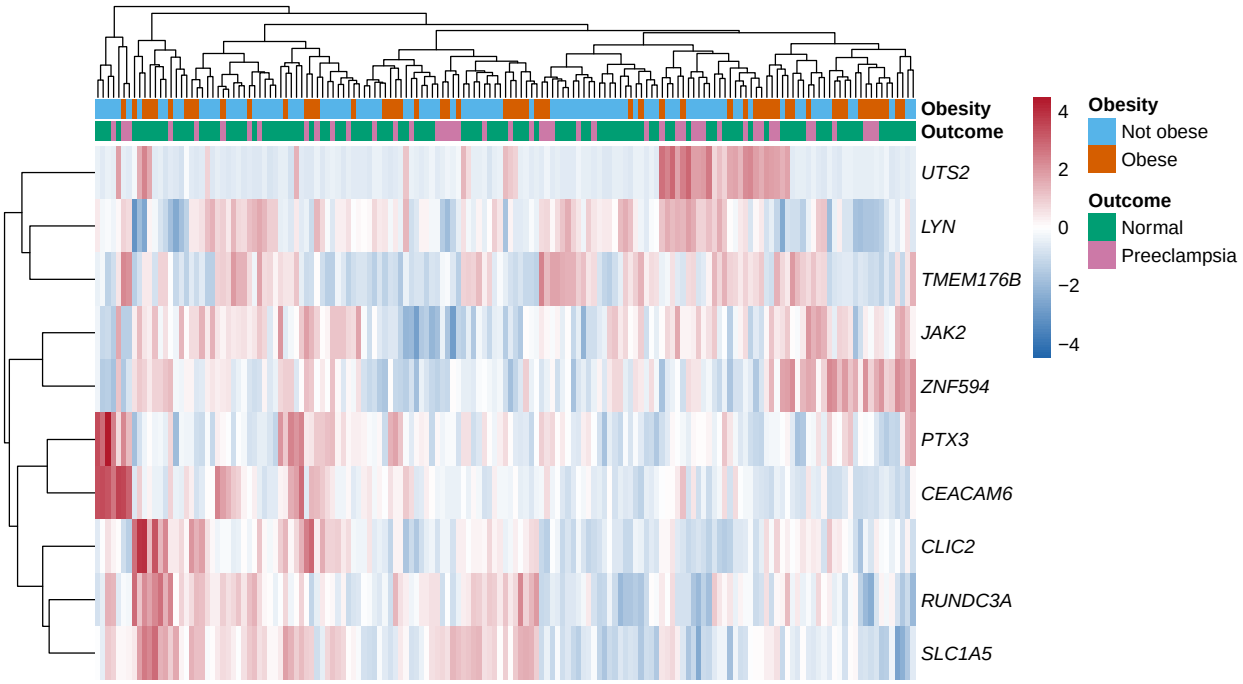


Figure 2: Heatmap of ten genes (*LYN*, *JAK2*, *PTX3* and the seven most variable genes in the data set) in 157 pregnant women. Each column is one woman and each row is one gene. Colors show row-scaled expression (z-score of $\log_2$ intensity; red is higher and blue is lower than that gene's mean). Genes and women were clustered by hierarchical clustering (Euclidean distance, complete linkage). Tracking bars show pregnancy outcome and obesity status (not obese, BMI < 30 kg/m^2; obese, BMI ≥ 30 kg/m^2).

2.4 Mean expression by pregnancy outcome

Compared with normal pregnancies, mean expression in preeclampsia was most increased for *UTS2* (difference in mean z-score, 0.21) and most decreased for *RUNDC3A* (-0.38). All differences were smaller than 0.4 standard deviations, indicating only modest differences between pregnancy outcomes for these genes (Figure 3).

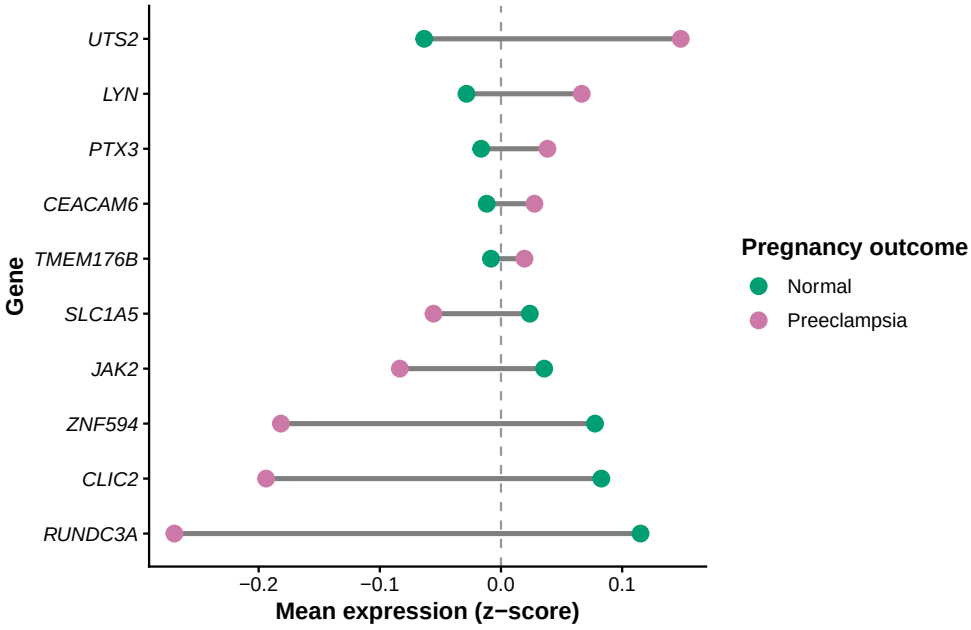


Figure 3: Dumbbell plot of mean expression of the ten heatmap genes by pregnancy outcome. Expression of each gene was converted to a z-score across all 157 women; points show the mean z-score in normal pregnancies (green) and preeclampsia (pink), and the grey line shows the difference between groups. Genes are ordered from the largest increase (top) to the largest decrease (bottom) in preeclampsia.

References

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